

Li Chuin Chong



Computational Virologist

Ph.D. Candidate • TWINCORE GmbH c/o HZI • Hannover Medical School
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SUMMARY

Computational virologist working at the interface of viral evolution, large-scale sequence analysis, and data-driven approaches in computational biology. My research focuses on understanding viral diversity and emergence using high-throughput sequencing data. I develop scalable computational approaches for virus discovery, diversity analysis, and spillover risk assessment, with a particular interest in quantitative and integrative frameworks for studying viral evolution.

Currently, I am a PhD candidate in Biomedical Data Science at Hannover Medical School (TWINCORE), where my work is associated with the COVIPA consortium focused on understanding virological and immunological drivers of COVID-19 pathogenesis.

Virus evolution • Virus discovery • Spillover risk assessment

CURRENT APPOINTMENTS

TWINCORE Centre for Experimental and Clinical Infection Research c/o
Helmholtz Centre for Infection Research (HZI) | Hannover, DE
Institute for Experimental Virology

Doctoral Researcher

Apr 2022 - Present

EDUCATION

Hannover Medical School | Hannover Biomedical Research School, Germany

Ph.D., Biomedical Data Science

Apr 2022 - Present

- Project title: Increasing pandemic preparedness by computational high-throughput virus discovery: Identification of RNA viruses and host reservoirs with high spillover risk
- Supervisory committee: Chris Lauber (main), Stefan Seitz, Alice McHardy, Daniel P. Depledge

Perdana University | School of Data Sciences, Malaysia

MRes, Bioinformatics

2020

- Dissertation: Mapping the minimal set of the viral peptidome
- Awarded with Distinction (the highest)
- Supervisory committee: Asif M. Khan (main), Sy Bing Choi, Kenneth Ban Hon Kim

Universiti Putra Malaysia | Faculty of Medicine and Health Sciences, Malaysia

B.Sc., Biomedical Sciences

2018

- Dissertation: Expression, purification and characterisation of the protruding domain of *Macrobrachium rosenbergii* nodavirus capsid protein
- Supervisory committee: Kok Lian Ho, Abdah MD Akim

PROFESSIONAL EXPERIENCE

Doctoral Researcher

Apr 2022 - Present

Institute for Experimental Virology, TWINCORE Centre for Experimental and Clinical Infection Research c/o Helmholtz Centre for Infection Research (HZI), Germany

Visiting Trainee

Oct - Nov 2023

Division of Virus Associated Carcinogenesis (F170), German Cancer Research Center (DKFZ), Germany

Research Intern

May 2020 - Dec 2021

Faculty of Medical Technology, Mahidol University, Thailand

International Visiting Researcher

Feb - Sep 2021

Beykoz Institute of Life Sciences and Biotechnology (BILSAB), Bezmialem Vakif University, Turkey

Research Assistant

Sep - Dec 2020

Faculty of Dentistry, University of Malaya, Malaysia & Cancer Research Malaysia

Research Assistant

Dec 2019 - May 2020

Perdana University School of Data Sciences (PU-ScDS), Malaysia

SKILLS**Experimental Techniques (previously used)**

Molecular and structural biology: DNA/plasmid extraction, PCR, protein expression and purification, electrophoresis, western blotting, chromatography (IMAC, FPLC), protein characterisation (CD, DLS)

In vivo techniques: Laboratory animal handling, dissection, and phlebotomy

Bioinformatics

Comparative genomics & Phylogenetics: Multiple sequence alignment, phylogenetic analysis, sequence clustering, evolutionary analysis

Vaccine informatics & Immunoinformatics: Entropy-based quantitative analysis, motif identification analysis, diversity analysis, immune-relevance analysis

Structural bioinformatics: Protein and RNA structure annotation, structural visualisation (PyMOL, UCSF Chimera, VMD), secondary structure prediction, protein structure analysis, protein-protein docking and interaction analysis

Viral informatics & Metagenomics: Viral genome annotation, SRA-based virus discovery, metagenomic virome profiling (Taxonomic characterization of viromes, Abundance estimation of viral population), host-pathogen interaction analysis

High performance computing (HPC) environments: PBS, SLURM

Workflow automation and management: Snakemake

Data Science

Exploratory data analysis, data visualisation, biostatistics, web scraping, application development

Web Development

HTML, CSS, JavaScript

Programming Languages (intermediate)

Python, R, shell scripting, GitHub CI/CD pipeline

PUBLICATIONS

Citations: 395 | **h-index:** 11 | **Papers Published:** 25

For a complete list of publications, please visit my [Google Scholar profile](#).

Chong et al. Quantifying Zoonotic Spillover Risk: Identifying High Risk RNA Viruses with PREDICTORix. *Manuscript in preparation*.

(selected; *co-first author)

Chuprikova, L., **Chong, L. C.***, Ruff, S., Seitz, S., & Lauber, C. (2026). Data-driven virus discovery with virushunter and virusgatherer. *bioRxiv*. doi:[10.64898/2026.01.26.701714](#).

- Chong, L. C., & Khan, A. M.** (2024). A systematic bioinformatics approach for mapping the minimal set of a viral peptidome. *Current Protocols*. doi:<https://doi.org/10.1002/cpz1.1056>.
- Chong, L. C., & Lauber, C.** (2023). Viroid-like RNA-dependent RNA polymerase-encoding ambiviruses are abundant in complex fungi. *Frontiers in Microbiology*, 14. doi:<https://doi.org/10.3389/fmicb.2023.1144003>.
- Chong, L. C., & Khan, A. M.** (2022). Historical milestone in 42 years of viral sequencing — impetus for a community-driven sequencing of global priority pathogens. *Frontiers in Microbiology*, 13. doi:<https://doi.org/10.3389/fmicb.2022.1020148>.
- Chong, L. C., & Khan, A. M.** (2022). Negligible peptidome diversity of SARS-CoV-2 and its higher taxonomic ranks. *bioRxiv*. doi:<https://doi.org/10.1101/2022.10.31.513750>.
- Chong, L. C., & Khan, A. M.** (2019). Identification of serotype-specific, highly conserved dengue virus sequences: Implications for vaccine design. *BMC Genomics*, 20(Suppl 9), 921. doi:<https://doi.org/10.1186/s12864-019-6311-z>.

BOOK CHAPTERS

- Chong, L. C., & Khan, A. M.** (2025). Vaccine target discovery. *Encyclopedia of Bioinformatics and Computational Biology, 2nd Edition*. doi:[10.1016/B978-0-323-95502-7.00226-8](https://doi.org/10.1016/B978-0-323-95502-7.00226-8).
- Chong, L. C., & Khan, A. M.** (2019). Vaccine target discovery. *Encyclopedia of Bioinformatics and Computational Biology*. doi:[10.1016/B978-0-12-809633-8.20100-3](https://doi.org/10.1016/B978-0-12-809633-8.20100-3).

SOFTWARE

PREDICTORix: A scalable, phylogeny-aware pipeline for quantifying RNA virus spillover risk

 [PREDICTORix \(private mode\)](#) • Developer (main)

UNIQmin: An alignment-independent tool for the study of pathogen sequence diversity at any given rank of taxonomy lineage

 [PyPi](#) •  [UNIQmin](#) • Developer (main) and maintainer

vDiveR: A graphical user interface (GUI)-based web application hosted on R Shiny for the visualization of various diversity dynamics

 [CRAN](#) •  [vDiveR](#) • Co-developer

VirusHunterGatherer: A Snakemake pipeline for data-driven virus discovery

 [VirusHunterGatherer](#) • Contributor

Virusparies: An R package designed for visualizing outputs from VirusHunterGatherer

 [Virusparies](#) • Contributor

GRANTS

Intramural Funding

Helmholtz Visiting Researcher Grant (€4,000)	2023
Role: Visiting Trainee	

Extramural Funding

PUTI Q1 Grant from Universitas Indonesia (IDR 120 million)	2022
Role: Collaborator PI: Toar Jean Maurice Lalisang (main) & Linda Erlina (co-)	
Malaysian Medical Association (MMA) (MYR 5,000)	2019
Role: Collaborator PI: Mohammad Asif Khan (main) & Choi Sy Bing (co-)	

TEACHING AND SERVICE

Graduate Student Supervisor

Supervisor

Stephen Sugiharto • B.Sc. (Bioinformatics), I3L, ID 2022-2023

Student Mentor

Pendy Tok • MRes (Bioinformatics), UM, MY 2021-Present

Sergej Ruff • M.Sc. (Biomedical Data Science), MHH, DE 2024

Tham Hong Yun • Ph.D. (Microbiology), UPM, MY 2018-2024

Sofiah binti Tajuddin • MRes (Microbiology), UPM, MY 2019-2022

Interns and student volunteers at BVU • Beykoz Institute of Life Sciences and Biotechnology (BILSAB), TR 2021

Srosha Khanjou • B.ASc (Medicine), MUIC, TH 2020-2021

Interns at CaRE UPM • B.Sc. (Biomedical Sciences), UPM, MY 2017-2018
• B.Sc. (Biomedical Sciences), UM, MY 2018

Outreach & Professional Societies

Leadership

- Finance Committee Member, ISCB Student Council (ISCBSC) 2023-Present
- Organizing Committee Member, APBioNETTalks 2020-2023
- Ambassador, Asia Pacific Bioinformatics Network (APBioNET) 2021-2022
- Public Relations & Outreach (PRO) Committee Member, Global Organisation for Bioinformatics Learning, Education & Training (GOBLET) 2019-2021
- Executive Council of Association for Medical and Bio-Informatics Singapore (AMBIS) 2018-2019
- Executive Council of Alumni and Leadership of Biomedical Science Club, UPM 2015

PRESENTATIONS

Keynote/Invited talks

An overview of the RNA virus discovery field since the 2023 RdRp Summit!

11-12 May 2025

- RdRp Summit 2025, Lisbon, Portugal

Unravelling the virosphere: Insights from viral discovery and sequence diversity analysis

17 Jun 2023

- Guest Lecture series of Indonesian Medical Education and Research Institute (IMERI)

Contributed talks

PREDICTORix: A scalable phylogeny-aware approach for quantifying viral spillover risk

11-12 May 2025

- RdRp Summit 2025, Lisbon, Portugal

A near-full compression of SARS-CoV-2 peptidome using UNIQmin

12-21 Sep 2022

- 21st European Conference on Computational Biology (ECCB2022), Stiges, Barcelona

Identification of highly conserved, serotype-specific dengue virus sequences: Implications for vaccine design

10-12 Sep 2019

- 18th International Conference on Bioinformatics (InCoB 2019), Jakarta, Indonesia

Posters (*selected*)

PREDICTORix: A scalable phylogeny-aware approach for quantifying viral spillover risk

13-15 May 2025

- International Virus Bioinformatics Meeting (ViBioM 2025), Lisbon, Portugal

Towards quantifying zoonotic spillover risks of RNA viruses: A comprehensive approach using next-generation sequencing data

16-20 Sep 2024

- 23rd European Conference on Computational Biology (ECCB 2024), Turku, Finland

UNIQmin, an alignment-independent tool for the study of viral sequence diversity at any given rank of taxonomy lineage: HPC solutions for big-data performance boost

30 Aug-3 Sep 2021

- 2021 ACM Europe Summer School (virtual)

Mapping the minimal set of the viral peptidome across all major viral taxonomies

25-30 Jul 2021

- 29th Conference on Intelligent Systems for Molecular Biology and 20th European Conference on Computational Biology (ISMB/ECCB 2021; virtual)

HONORS AND AWARDS

(*selected*)

- MHH Congress Scholarship & ISME Travel Fellowship – Oral & Poster presentation, 2nd RdRp Summit & ViBioM2025
- Best Project Award, Collaborative Bioinformatics Hackathon 2025: Machine learning and AI approaches to multimodal problems in computational biology
- OBF Travel Fellowship – Poster presentation, ECCB 2024
- GOBLET Fund & Malaysia Trust Fund Travel Fellowship – Oral presentation, ECCB 2022
- First Prize, BCM/HGSC Virtual Hackathon 2021: Pan-structural Variation in the Cloud
- ESCV Fellowship – Poster presentation, ESCV 2021
- ISCB Fellowship – Poster presentation, ISMB/ECCB 2021
- Best E-Poster Award, ICBCBBS 2021
- APBioNET Travel Fellowship – Oral presentation, InCoB 2019
- Public Service Department - National Degree Programme (JPA-PIDN) Scholarship for B.Sc. (Biomedical Sciences), 2015-2018

REFERENCES

Dr. Chris Lauber

Head of Computational Virology
Institute of Experimental Virology,
TWINCORE GmbH
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RESIST Cluster of Excellence, Medizinische
Hochschule Hannover (MHH)
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